Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [14CE45]

FOURTH SEMESTER B.Tech. EXAMINATION, JUNE – 2016

SOIL MECHANICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. Answer any **FIVE** full Questions choosing at least any two full questions from each part.

ii. Assume any missing data suitably if necessary

iii. Draw neat diagrams wherever required.

PART - A

1. a. Describe briefly sieve analysis of soils. Define coefficient of curvature and coefficient of uniformity for well graded soil with the help of typical graph. **05**

   b. A soil specimen has a water content of 10% and a wet unit weight of 20 kv/m^3. If the specific gravity of solids is 2.7, determine the dry unit weight void ratio, and the degree of saturation. Take γ_w=10kN/m^3. **05**

   c. Give a brief note on plasticity chart to classify the fine grained soil. **04**

2. a. Derive capillarity in soil. And drive an expression for capillary rise. **07**

   b. A sand deposit is 10m thick and overlay a bed of soft clay. The ground water table is 3m below the ground surface. If the sand above the ground water table has a degree of saturation of 45%, plot the diagram showing the variation of the total stress, pore water pressure and the effective stress. The void ratio of the sand is 0.7 take G=2.65. **07**

3. a. List out the methods for determination of coefficient of permeability. **04**

   b. Write a note an seepage and quick sand condition. **05**

   c. What is a flow net? Explain it with a neat sketch. **05**

4. a. How do you calculate stresses due to line load and circular load? **07**

   b. A concentrated load of 2000kN is applied at the ground surface. Determine the vertical stress at a point which is 6m directly below the load. Also calculate the vertical stress at a point which is 5m away from the axis of the load. Also compare with the Westergaard's value. **07**

PART - B

5. a. List out the 3 major differences between Standard Proctor Compaction Test and Modified Proctor Compaction Test. **03**

   b. Write a note on Proctor's needle method for determination of field water content. **05**

c. The following results were obtained from SPCT on a sample of soil. **06**

w.c%	0.12	0.14	0.16	0.18	0.2	0.22
Mass of wet soil(kg)	1.68	1.85	1.91	1.87	1.87	1.85

The volume of the mould is 950ml. Plot the compaction curve and obtain the max dry density and OMC.

6. **a.** What are the assumptions and limitations of Terzaghi's theory? **03**

b. Give definitions of normally consolidated, under consolidated and over consolidated soils. **03**

c. Calculate the final settlement of the clay layer shown in fig 6(c) due to an increase of pressure of 30kN/m^2 at mid-height of the layer. Take γ_w=10kN/m^3. Also calculate the settlement when the water table rises to the ground surface. **06**

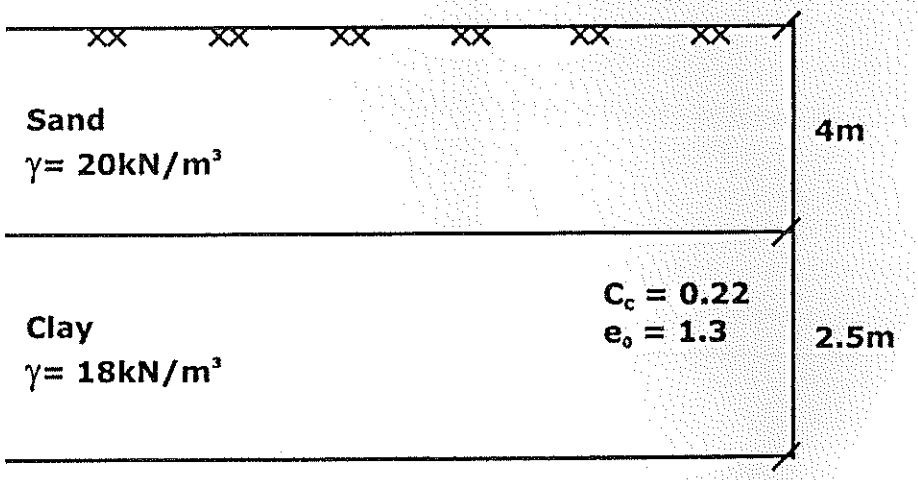

Fig. 6(c)

7. **a.** Write a detailed note an Mohr-coulomb failure theory. **06**

b. List out the factors affecting shear strength of sands and clays. **08**

8. **a.** Write a note on Direct Shear Test. **06**

b. List out the methods based on the drainage and explain any one method with a neat sketch and graph. **08**